Design note: the Tier-3 ASTRA MLP emulator — dataset, splits, and diagnostics

ASTRA clustering project

June 2026

Abstract

This note specifies the first MLP emulator built on the Tier-3 pilot (`data/fullbox_tier3/`: 10 cosmologies × 50 HODs, all complete). It fixes the emulator *inputs*, the *target* data vector (prioritising the void/knot environments from the randoms, then the data, in monopole and quadrupole), the *cached dataset* that decouples slow disk reads from fast retraining, the *train/validation/test split* (leave-one-cosmology-out, not a random split), the *architecture*, and the *diagnostic plots*. It is the companion to `tier3_emulator_note.tex` (which sets the campaign strategy); this note is the operational recipe the code implements.

1 Inputs (20-D)

The emulator maps $\theta = (\theta_{\rm cosmo}, \theta_{\rm HOD}) \mapsto \xi(s)$. The Tier-3 block `c130–c181` varies the broader $w_0 w_a$CDM+ set, so the cosmology input is **8 varying parameters** (read from `data/abacus_cosmologies_params.`

$$\omega_b,\ \omega_{\rm cdm},\ h,\ n_s,\ \alpha_s,\ N_{\rm ur},\ w_0,\ w_a.$$

(A_s and $\omega_{\rm ncdm}$ are fixed across the block; σ_8 is a derived quantity and is kept as a label, not an input, to avoid collinearity.) The HOD input is the **12 yuan23 parameters** already used throughout the project: `LOGM_CUT, LOGM1, SIGMA, ALPHA, KAPPA, ALPHA_C, ALPHA_S, S, ACENT, ASAT, BCENT, BSAT`. Total input dimension **20**. All inputs are standardised (zero mean, unit variance over the training set).

2 Targets

The science priority is the **extreme environments** — voids (Q1) and knots (Q4) — **mostly from the randoms**, then from the data, in both multipoles. The primary target legs are therefore

$$\{\texttt{rand\_q1},\ \texttt{rand\_q4},\ \texttt{cross\_full\_rand\_q1},\ \texttt{cross\_full\_rand\_q4}\}$$

followed by the data twins {`data_q1`, `data_q4`, `cross_full_data_q1`, `cross_full_data_q4`}, each in $\ell = 0$ and $\ell = 2$. The current pilot multipoles are stored on the coarse 15-bin 0–150 h^{-1} Mpc grid ($\sim 10\,h^{-1}$ Mpc bins); the fine small-scale binning the strategy note calls for would require a pipeline re-run and is deferred. We train on the bins on disk and report accuracy as a function of a scale cut $s_{\rm cut}$ so the small-scale ($s < 40\,h^{-1}$ Mpc) payoff region is read off directly.

3 Cached dataset — the key to cheap retraining

We will retrain many times (different target subsets, architectures, PCA truncations, loss weights). Re-globbing ∼1000 run directories each time is wasteful, so a one-shot builder (`scripts/build_emulator_dataset.` writes a single compact archive `data/emulator_tier3/dataset.npz` holding the *full* stem set — all 17 stems $\times$ 2 multipoles $\times$ 15 bins (510-D) — even though the first model trains on the void/knot subset. Arrays:

- `X_cosmo` $(N, 8)$, `X_hod` $(N, 12)$ — standardisable inputs;
- `Y` $(N, 510)$ — concatenated $[\xi_0, \xi_2]$ over stems;
- `Ynoise` $(N, 510)$ — per-bin ASTRA-iteration scatter (`xi*_std`);
- `s` (15,), `stem_labels`, `ell_labels`, `bin_index` — block bookkeeping;
- `cosmo_id`, `hod_id`, `sigma8` — per-row metadata for grouping/plots.

Target selection at train time is a column mask over `Y`; no disk re-read. The same builder writes `dataset_anchor.npz` from the 450 Fisher-grid runs (`data/fullbox/`, cosmologies `c000,c100--c105,c112,c113`), which are *not* in the training block and serve as an independent generalisation anchor in the LCDM neighbourhood.

4 Train / validation / test split

The decisive structural fact: there are only **10 cosmologies** but 500 runs. A random split over runs would place HODs of the same cosmology in both train and test, leaking the (sparse, hard) cosmology axis and giving an optimistic score. In deployment the emulator predicts at *new* cosmologies, so the split is made **along the cosmology axis**:

- **Headline: 10-fold leave-one-cosmology-out (LOCO).** Each fold trains on 9 cosmologies (450 runs) and tests on the held-out one (50 runs); rotating over folds tests every cosmology. This is the go/no-go metric.
- **Inner validation:** within each fold's 9 training cosmologies, ∼15% of the HODs (drawn across all 9, fixed seed) are held out for early-stopping and hyperparameter choices.
- **Per-fold fractions** ≈ 76% train / 14% validation / 10% test, partitioned by cosmology — *not* a random 70/15/15 over runs.
- **External anchor:** the 450 Fisher-grid runs, never trained on, give a second test set in the dense LCDM region we ultimately care about.

LOCO is chosen over a fixed permanent holdout because 10 cosmologies is too few to also sacrifice a standalone test block.

5 Architecture

SUNBIRD-style feed-forward MLP in `torch` (2.10, CUDA available):

- Input 20-D (standardised) → 3–4 hidden layers (256–512 units, SiLU) → output;

- **Output via PCA** ($\sim$10–20 components on the standardised target vector) for a compact, decorrelated regression target;
- **Noise-weighted MSE** loss, weight $1/\sigma^2_{\rm noise}$ per bin, for the pilot. (The covariance-weighted loss that matches the inference metric is the eventual goal, but the covariance is the known weak point of Tier-3, so we start diagonal.)
- **Ensemble** of $\sim$5 independently-seeded MLPs for the emulator uncertainty (used in the pull/calibration diagnostic);
- Adam, early stopping on the inner validation loss.

6 Diagnostic plots

Acceptance is judged per s-bin against two yardsticks already used at Tier-2: the **ASTRA noise floor** (mean per-iteration $\xi_{\rm std}$, which no emulator should be expected to beat) and the **signal spread** (std of the training vectors, i.e. what there is to learn). A good emulator has LOCO RMS $\sim$ noise floor and $\ll$ spread, especially at $s < 40\,h^{-1}\,$Mpc. Plots:

1. **Learning curves** — train/validation loss vs epoch, per fold (overfit check).
2. **LOCO prediction vs truth** — $s^2\xi$ for void/knot, data & random, $\ell = 0, 2$, on the held-out cosmology.
3. **Error vs yardstick** — per-bin LOCO RMS against noise floor and signal spread vs s, zoomed to $s < 40\,h^{-1}\,$Mpc (the acceptance plot).
4. **Pull histograms** — $(\hat{\xi} - \xi)/\sigma_{\rm ens}$ should be $\mathcal{N}(0, 1)$ (ensemble-uncertainty calibration).
5. **Accuracy vs scale cut** — RMS/noise integrated for $s < s_{\rm cut}$, directly answering "sub-noise at $s < 40$".
6. **Fractional-error heatmap** — (stem $\times\,\ell$) vs s-bin, colour = median |frac. error|; an at-a-glance map (expect random quadrupoles weakest, per prior findings).
7. **External-anchor prediction vs truth** — the same curves on the 9 Fisher cosmologies.

7 Caveats

Ten cosmologies over an 8-D cosmology input is thin: a poor LOCO result means "need the full `c130`–`c181`×120 campaign," not "ASTRA fails." The coarse 15-bin grid limits the very small-scale resolution that carries most of the Fisher signal. The HOD response is assumed smooth and the ASTRA-iteration noise (3 iterations in the pilot) sets a floor the emulator cannot cross. These are the questions the diagnostics are designed to expose.